RiboSurfBind

(A) RNA inputs

RNA sequence

G G C C A G A U

ERNIE-RNA
Frozen encoder

RNA structure

Ligand-free RNA

(B) Semantic–spatial residue learning

Attention

Semantic top-k = 10

768 → 128
Initial states $H^{(0)}$

Spatial · kNN = 8

Initial residual connection · $H^{(0)}$

Layer 1

Gated update

Layer 2

Gated update

$H^{(0)}$ $H^{(1)}$ $H^{(2)}$ Layers 3–4 $H^{(4)}$

N × 128

(C) Molecular-surface learning

Coordinates & local frames

RNA surface
Ligand-free

Point features
29D

Surface encoder
Quasi-geodesic blocks × 4

M × 64

Surface points → residue

Point features

Max

Attention

Assigned to i
64D each

Residue i

64D ‖ 64D
128D feature

N × 128

(D) Residue-wise relation gating

Layer ℓ · residue i

Semantic message $m_{s,i}$

Gate inputs
x_i, $m_{s,i}$, $m_{p,i}$

Spatial message $m_{p,i}$

Relation gate
MLP → Softmax

$w_{s,i}$

$w_{p,i}$

m_i

To GCNII

$w_{s,i} + w_{p,i} = 1$

(E) Fusion and prediction

Separate LN

Residue branch

Surface branch

128D each

Concatenate

256D

MLP

1 128 256

Binding

Non-binding

Sigmoid
+ decision

M: surface points; N: residues. Graphs and feature colors are schematic.